PROVIDER CREDIBILITY PREDICTION SYSTEM

INDIVIDUAL THESIS

Service Provider Platform

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Individual component

Document Extraction and Provider Credibility Scoring

R26-IT-072

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science Honours in Information Technology

Specialized in Information Technology

FACULTY OF COMPUTING

Sri Lanka Institute of Information Technology

Sri Lanka

2026 OCTOBER

DECLARATION AND SUPERVISOR CERTIFICATION

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Abstract

This thesis presents a document-supported provider credibility scoring component for the PAWAN service-provider platform. The component transforms identity, registration, qualification, experience and portfolio evidence into structured inputs, predicts a numerical score and returns the result to a provider database. A synthetic dataset of 2,000 records was generated using a capped weighted scoring policy. Random Forest and XGBoost regressors were tuned with three-fold cross-validation and evaluated using a seed-42 split containing 1,600 training and 400 holdout records. The recorded Random Forest result was RMSE 1.569 and R squared 0.989; XGBoost achieved RMSE 1.141 and R squared 0.994 and was selected for inference. Native PDF extraction, optional OCR and rule-based feature interpretation produced local demonstration results for four example profiles. The integration notebook records one provider update with a score of 84.60 and an Intermediate level. Supplementary offline analysis confirmed that the target formula exactly reproduces every dataset label, while identity is constant and 33.80% of records reach the maximum score. The implementation therefore demonstrates policy approximation and a document-to-database prototype rather than independently validated prediction of provider reliability. Document-presence fallbacks, inconsistent category normalisation and absent extraction and stakeholder benchmarks limit verification claims. The individual contribution is the connected dataset, modelling, extraction and integration workflow and its evaluation. Future work should introduce explicit review states, consistent preprocessing, a representative document benchmark and independently justified credibility labels before operational deployment.

Keywords: provider credibility, document extraction, XGBoost, synthetic data, service marketplace

ACKNOWLEDGEMENTS

I acknowledge the supervision and feedback provided by [Supervisor name] and [Co-supervisor name if applicable] during this individual research component. I thank [Group members or contributors] for the specific assistance they provided with [Describe confirmed assistance].

I acknowledge the use of open-source software and the research publications cited in this thesis. Generative AI assistance in document preparation and supplementary analysis is disclosed in Appendix B.

[Confirm and complete this acknowledgement with the actual assistance received. State any funding or substantial technical support. Remove this instruction before submission.]

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LIST OF ABBREVIATIONS

AC Air conditioning

AI Artificial intelligence

API Application programming interface

BR Business registration

CSV Comma separated values

CV Cross-validation

ML Machine learning

NIC National Identity Card

NVQ National Vocational Qualification

OCR Optical character recognition

PDF Portable Document Format

RF Random Forest

RMSE Root mean squared error

RTDB Realtime Database

SDG Sustainable Development Goal

SHAP SHapley Additive exPlanations

SLIIT Sri Lanka Institute of Information Technology

XGBoost Extreme Gradient Boosting

1 INTRODUCTION

1.1 Overall Project Context

Service marketplaces need a practical way to present evidence about the people who offer services. A customer selecting an electrician, plumber or carpenter may see a provider profile before having any direct experience of that provider's work. At this point, information about qualifications, identity, previous employment and completed projects can help the customer understand the available evidence. A numerical summary can support this process if its meaning and limitations are clear.

The PAWAN project context for this individual component is a service-provider platform that stores provider records and uploaded documents. The implementation connects a Python processing pipeline to provider data in Firebase Realtime Database. The individual component concerns document processing and credibility scoring. Booking, payments, customer reviews and the implementation of other group members' components are outside the demonstrated scope of this component.

The platform context creates an onboarding problem. Providers may submit different document types and file formats, while the application needs structured fields that can be processed consistently. A certificate is a document, but a prediction model expects a count, a recognised qualification category and a numerical representation. Experience letters similarly contain text that must be converted into a duration. The component provides the conversion layer between uploaded evidence and a structured provider assessment.

The term credibility in this thesis refers to the prototype score defined by the project. It combines document-derived indicators and professional background features. It does not mean that the system can certify a person's honesty, legal status or future service quality. A higher score means closer alignment with the project's scoring policy. Establishing that this policy predicts real customer outcomes requires a separate study.

1.2 Individual Component Context

The individual component is the Provider Credibility Prediction System. It contains four connected activities: generating structured experimental data, comparing regression models, extracting model inputs from documents and returning results to the provider database. The training code evaluates Random Forest and XGBoost. The inference code produces a score between zero and one hundred and assigns Beginner, Intermediate or Professional as the corresponding presentation category [11]-[14].

The document pipeline handles identity documents, business-registration documents, certificates, experience evidence and portfolios. Portable Document Format (PDF) files with embedded text are processed using Poppler's text-extraction command. Images and image-based PDFs can be processed using optical character recognition (OCR). Extracted text is interpreted by keyword and regular-expression rules before the model receives an ordered feature vector.

The database integration is implemented separately from local inference. It retrieves provider records, selects records with pending documents, downloads their document files, extracts features and writes a structured result. This separation allows the model and document logic to be examined without depending on a live database during every experiment. It also creates two extraction implementations whose assumptions need to remain consistent.

1.3 Problem Statement

The problem addressed is how to convert heterogeneous provider documents into a consistent, inspectable numerical assessment during provider onboarding. Manual inspection requires a reviewer to read each document and interpret its relevance. A system that records only the presence of uploaded files provides less information than a system that extracts qualification level, experience duration and evidence counts. Conversely, an automated score can mislead users when uncertain extraction is presented as successful verification.

The research problem therefore has both a modelling dimension and an evidence dimension. The modelling dimension asks whether a regression model can reproduce

a defined scoring policy from structured features. The evidence dimension asks whether documents can be converted into those features and connected to the platform record. Good regression performance alone cannot answer the second question because the model is evaluated using already structured data.

The prototype must also handle providers with limited evidence. A missing document, an unreadable document and a document that fails authenticity checks are different conditions. The current implementation does not fully distinguish them. Its behaviour is examined throughout the thesis because this distinction affects the interpretation of the resulting score and verification flags.

1.4 Research Gap

Reputation research provides a basis for understanding how information about previous interactions can influence decisions between strangers [1]. The present component addresses the earlier onboarding stage, when a provider may have documents but little platform transaction history. Its practical gap is the connection between submitted documents, a structured provider feature schema and a score that can be stored in the service platform.

The project does not establish that existing research has never combined document analysis and scoring. Instead, the gap is defined within the application requirements: the platform needs an implemented workflow for the particular document groups, service categories and database structure used by this project. This is an application and integration gap. A claim of a new general credibility theory would require evidence beyond the experiments described here.

There is a further evaluation gap. The prototype uses synthetic targets generated from a known weighted formula. Model evaluation on these targets tests approximation of that formula rather than independent judgements of credibility. The study explicitly separates this technical benchmark from future validation against experts, authenticated records or observed service outcomes.

1.5 Aim and Objectives

The aim is to develop and evaluate a document-supported provider scoring prototype that transforms submitted evidence into structured features, predicts a credibility score and connects the result to the provider platform.

The individual objectives are to define a provider feature schema and synthetic experimental dataset; compare Random Forest and XGBoost regression using a reproducible split and cross-validation; implement PDF and image text extraction with rules for deriving model features; present the predicted score through defined credibility levels; integrate feature extraction and scoring with Firebase provider records; and identify the limits of the prototype through software inspection and result analysis.

The objectives distinguish implemented functions from validation claims. Producing a working prediction function is a software objective. Demonstrating that a score reflects actual service reliability is a research validation objective that remains open. The conclusion evaluates each objective against the available implementation and experimental evidence.

1.6 Research Questions

RQ1 asks which of Random Forest and XGBoost achieves lower error when approximating the project's synthetic credibility scores on the selected holdout set. This question is assessed using root mean squared error (RMSE) and the coefficient of determination, R squared.

RQ2 asks whether the prototype can convert example provider documents into the expected structured features and score categories. The recorded local demonstrations provide evidence that the workflow executes on the selected examples. They do not supply a labelled extraction benchmark or a measured OCR accuracy rate.

RQ3 asks whether extracted features and a credibility result can be formatted and returned to the provider's Firebase record. The integration notebook records one successful update. RQ4 asks what assumptions constrain the interpretation and

generalisation of the score. This is answered through examination of label generation, feature variation, extraction rules and integration fallbacks.

1.7 Scope Limitations and Assumptions

The training dataset contains 2,000 simulated records across ten service categories. The regression input uses ten features. Identity verification is fixed at one in every generated record. Certificate and portfolio values are generated or assigned by project rules. These assumptions simplify prototype development but limit the conclusions that can be drawn about real providers.

The implemented document logic focuses on readable text and English duration expressions. The local file collector accepts PDF and selected image extensions. No video-analysis implementation is demonstrated, even though supporting documentation mentions videos as a possible portfolio input. The scanned-PDF fallback processes the first rasterised page. The work therefore does not establish complete multi-page or multilingual document coverage.

Authenticity checks against issuing authorities, biometric identity matching, document-tampering detection and customer outcome validation are outside the implemented scope. The software uses the feature name identity_verified, but its value reflects recognition rules or a document-presence fallback. This distinction is necessary to avoid attributing stronger verification capability to the component than its code supports.

1.8 Individual Contribution

The technical contribution described in this thesis consists of the dataset generator, model-comparison workflow, local document inference pipeline and Firebase integration pipeline contained in the individual component. The notebooks connect tabular modelling to document-derived features and provide an inspectable route from evidence files to platform outputs [11]-[14].

The research contribution is an evaluated prototype and an analysis of the boundary between policy approximation and credibility validation. The comparison identifies XGBoost as the better of the two recorded regression models on this synthetic

benchmark. The analysis also identifies weaknesses that matter to subsequent development, including constant identity inputs, saturation at the maximum score and fallbacks that treat upload presence as verification.

This attribution is limited to the supplied individual component. It does not assign ownership of the entire platform or describe other students' work. Any final group contribution statement should be reconciled with the group's agreed responsibilities and repository history.

1.9 Thesis Structure

Chapter 2 reviews reputation, tree ensembles, model explanation and document-understanding approaches. Chapter 3 explains the research design, dataset and evaluation procedure. Chapter 4 presents requirements and solution design. Chapter 5 describes implementation, integration and testing. Chapter 6 reports results with their evidence status. Chapter 7 discusses the findings and threats to validity. Chapter 8 examines commercialisation and sustainability as proposed directions. Chapter 9 concludes the individual work and identifies future development priorities. The appendices preserve supplementary technical details and an AI-assistance disclosure.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The review is a focused narrative review of sources directly relevant to the component. It covers reputation systems, regression ensembles, feature explanations, document text extraction and database integration. Original research papers and official software documentation are used for the technical foundations. Sources were selected for their connection to the implemented methods or a credible design alternative. This process is not presented as a systematic review with exhaustive database coverage.

The review separates three questions. The first concerns what information a platform uses to support trust. The second concerns how structured information is converted into a prediction. The third concerns how documents become structured information. Keeping these questions separate prevents a strong result in one layer from being treated as evidence for all layers.

The comparison uses methodological relevance rather than directly comparing numerical scores across unrelated datasets. Random Forest and XGBoost research explains why the methods are reasonable candidates. Document-understanding research explains alternatives to keywords and regular expressions. Reputation research explains the context in which a provider assessment may be useful. None of those sources validates the particular weights used in this project's target formula.

2.2 Domain and Theoretical Foundations

A provider profile contains several types of evidence. Identity information concerns who submitted the application. A business-registration document concerns an organisational record. A certificate concerns an educational or training claim. An experience letter concerns work history, and a portfolio concerns examples of work. These signals have different meanings and should retain their provenance when combined into a summary.

Resnick and Zeckhauser examine reputation information in online transactions, providing a foundation for the role of prior interaction histories in marketplace decisions [1]. In the PAWAN component, the initial inputs are documents rather than

a history of completed platform transactions. This means the score can support onboarding, while reputation developed through subsequent service interactions would be a separate source of evidence.

The target used in this project is operationally defined. A weighted sum gives points for selected features, and a cap constrains the total. The score is therefore a policy output. Learning this output with regression estimates the relationship specified by the generator. It does not independently discover which professional traits matter to customer satisfaction or safety.

This distinction is important for interpreting labels. Beginner, Intermediate and Professional are score bands defined by thresholds. They are not independently observed occupational classes. A provider crossing a threshold changes category because of a numerical rule. The category should therefore be accompanied by its definition rather than presented as an official qualification.

2.3 Related Research

Random Forest combines multiple randomised decision trees, with regression predictions aggregated across trees [2]. It is relevant because the provider inputs mix counts, binary indicators and numerical values. The project's comparison uses RandomForestRegressor as an ensemble baseline. The theoretical and methodological rationale comes from the algorithm, while its suitability for this dataset is established only by the experiment.

XGBoost is a tree-boosting system designed for efficient learning with regularisation [3]. Its sequential ensemble can fit relationships that include interactions and changes in response across feature ranges. The capped target formula gives the prototype such a response shape. The project uses XGBRegressor and compares its holdout predictions with those of Random Forest rather than assuming that boosting must perform better.

SHapley Additive exPlanations (SHAP) assigns feature contributions to a model prediction within an additive explanation framework [4]. The training notebook includes summary, dependence and individual explanation plots. These help inspect

the learned scoring behaviour. An explanation of a prediction does not establish that the input document is authentic or that a feature has a causal effect on reliability.

LayoutLM combines textual and layout information for document image understanding [6]. Donut investigates an OCR-free document-understanding approach [7]. They represent alternatives to a pipeline that first extracts text and then applies regular expressions. They were not trained or evaluated in this project. Their relevance is to future extraction improvements where layouts, templates and complex field relationships exceed the capabilities of the current rules.

The project consequently contributes an application prototype using established methods. It does not propose a new ensemble algorithm or claim benchmark superiority over document-understanding models. Its strongest comparison is between two models trained on the same structured experimental data.

2.4 Existing Systems Products and Techniques

Four approaches are useful design comparators: manual review, deterministic scoring, machine-learning regression and learned document understanding. Manual review can consider context and authenticity concerns, but requires review effort and a clear rubric. Deterministic scoring computes the project policy exactly and exposes its weights. Regression can approximate that policy, or later learn from independent outcomes if suitable labels become available.

The current implementation combines native PDF text extraction, Tesseract OCR and project-specific rules. Tesseract supports extracting printed text from images through command-line and programming interfaces [5]. The component then interprets that text using its own rules. OCR supplies text; it does not directly create an authenticated provider assessment.

GridSearchCV evaluates combinations of parameters using cross-validation [8]. The project applies three folds and R squared as the selection score. Firebase's administrative database interface supports updating selected keys and paths [9], which suits the integration payload containing result fields and status flags. These software

capabilities explain the implementation choices, while operational correctness depends on the application's validation and state handling.

Table 2.1 compares the approaches by the role they could play in this component. The comparison is conceptual; it is not a market survey or an empirical claim about named commercial services.

Table 2.1 Comparison of approaches for provider assessment

Approach	Main role	Strength	Constraint in this project
Manual document review	Interpret submitted evidence	Can examine context and contradictions	Requires reviewer time and a rubric
Deterministic weighted rule	Compute the current target policy	Exact and easy to inspect	Weights require independent justification
Random Forest or XGBoost	Predict a structured numerical target	Supports model comparison and later data learning	Current labels are synthetic
OCR with keyword rules	Convert documents into feature values	Inspectable and relatively simple	Sensitive to wording and extraction quality
LayoutLM or Donut	Learn document representations or extraction	Potentially handles more complex documents	Not implemented or evaluated here

2.5 Gap Synthesis and Conceptual Framework

The conceptual framework links submitted evidence to feature extraction, score estimation and platform presentation. A document must first be accessible and readable. Its contents must then be interpreted into a feature with a defined meaning. The feature must be encoded consistently with training. Finally, the model output must be interpreted according to the scope of its labels and displayed or stored without overstating certainty.

Errors can arise at each connection. An unreadable certificate may become an incorrect qualification value. An unknown service name may become a default category. A correctly calculated prediction may still be misleading if described as authority-level

verification. The framework therefore treats evidence provenance and uncertainty as part of assessment quality.

The application gap is addressed by implementing these connections for the project's schema and Firebase structure. The remaining research gap is the validation of the policy and extraction pipeline with independent evidence. The next meaningful study should compare the system with a deterministic rule and a human review rubric, then assess how extraction errors change the resulting score.

3 RESEARCH METHODOLOGY

3.1 Research Design

The study is organised as prototype development supported by a quantitative model comparison and a qualitative examination of implementation behaviour. The developed artefact links document processing, regression prediction and platform integration. The quantitative part measures prediction error on a synthetic dataset. The implementation analysis examines whether code and recorded demonstrations support the claimed functions.

The development sequence is dataset definition, categorical encoding, model tuning, holdout evaluation, model selection, local inference and database integration. This sequence reflects the notebooks in the component. The experimental unit for regression is a simulated provider record. The demonstration unit for document inference is a group of documents assigned to an example provider. These units must be kept separate because they support different conclusions.

The training notebooks preserve outputs from their original execution. Those outputs are reported as recorded results. Supplementary offline checks performed during thesis preparation analyse the CSV and isolated pure helper functions. They are identified separately and do not represent a repeat of the complete model-training or live integration experiment. No new connection to the provider database is required for the analysis in this thesis.

3.2 Data Participants and Study Materials

The primary modelling material is dataset/dataset.csv. It contains 2,000 records and thirteen columns: a provider identifier, ten predictor fields and two target-related fields. There are no recruited participants in the synthetic regression experiment. The generated records are experimental profiles rather than observations collected from actual service providers [11].

The generator sets the NumPy and Python random seeds to 42. It selects a service category from ten categories. Certificate count ranges from zero to three, reference count from zero to three and portfolio count from zero to five. Experience ranges from

0.5 to 15 years. All generated identity values are one. Business-registration values are sampled with a configured probability of 0.7 for registration; the realised dataset contains 1,426 registered records.

Qualification categories are None, NVQ Level 3, NVQ Level 4, Technical College and Vocational. When the certificate count is zero, the generator assigns None. Otherwise, it selects one of the other categories. Issuer-reputation values and portfolio-quality values are simulated numerical variables, not scores assigned by independent experts. Table 3.1 defines the predictor schema used by the notebooks.

Table 3.1 Predictor schema and interpretation

Predictor	Type	Interpretation in the prototype
service_category	Category	One of ten supported service categories
identity_verified	Binary	Identity indicator fixed at one in training
certification_count	Integer	Number of recognised certificates
highest_cert_level	Category	Certificate category selected by project rules
cert_issuer_reputation	Number	Simulated or heuristic issuer weight
business_registered	Binary	Registration indicator
experience_years	Number	Simulated or parsed years of experience
experience_reference_count	Integer	Count of experience evidence
portfolio_quality_score	Number	Simulated or heuristic portfolio indicator
portfolio_count	Integer	Number of portfolio evidence files

The supplementary document materials contain identity images and PDF examples for registration, certificates, experience and portfolios. The local inference notebook groups these materials into four demonstration profiles. Their filenames and example names are part of a test fixture arrangement, not a sampled study of provider

performance. The integration notebook also contains a recorded live-provider processing example. Personal identifiers are omitted from the thesis.

3.2.1 Target Generation

The numerical target is generated from a weighted sum and capped at one hundred. Equation 3.1 represents the implemented rule. The variables denote the identity indicator I, certificate count C, certificate-list index L, business-registration indicator B, experience years E and portfolio-quality value P.

$$S = min(100, 30I + 8C + 5L + 15B + 3E + 10P)$$

Equation 3.1 Synthetic target formula

The qualification index is taken from the generator's explicit list. It is a policy ordering, not an externally validated hierarchy of qualification value. Technical College has index three and Vocational has index four. The thesis preserves this rule when describing the experiment while treating its professional meaning as a limitation.

The target does not directly include service category, issuer reputation, experience-reference count or portfolio count. Those fields remain model inputs and can have correlations with other generated features. This means that model importance for a variable must be interpreted in the context of the generator rather than as independent evidence about real professional credibility.

3.2.2 Dataset Distribution

The realised dataset contains 1,254 Professional, 613 Intermediate and 133 Beginner records. These correspond to 62.70%, 30.65% and 6.65% of the dataset. A total of 676 records, or 33.80%, have the maximum score of one hundred. The score mean is 86.79585 and the median is 91.40. Table 3.2 and Figure 3.1 summarise this distribution.

Table 3.2 Realised target distribution

Score band	Record count	Percentage
Beginner	133	6.65

Score band	Record count	Percentage
Intermediate	613	30.65
Professional	1254	62.70
Score equal to 100	676	33.80

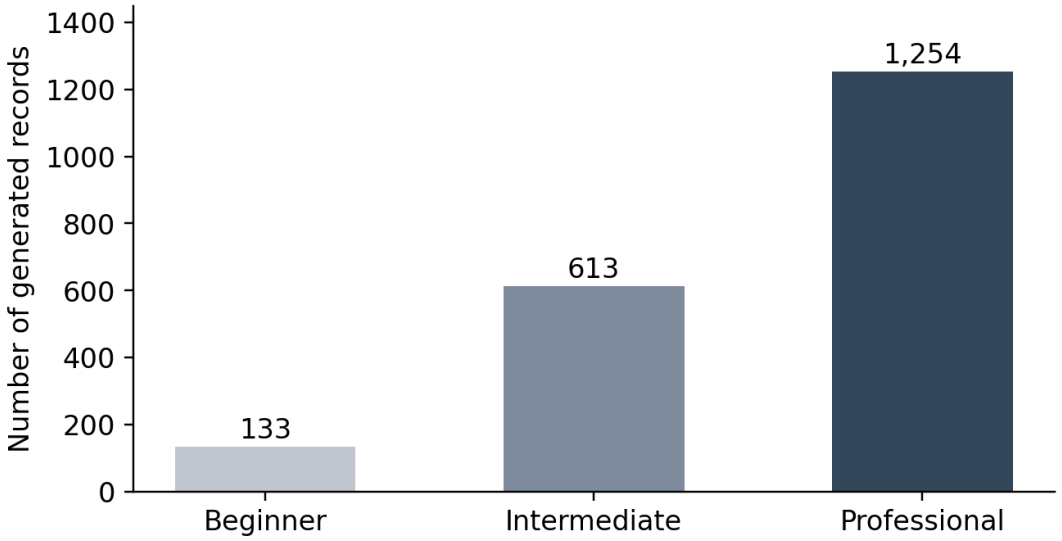


Figure 3.1 Credibility level distribution in the synthetic dataset

The score cap creates a concentration of records at the upper bound. Distinct feature combinations can therefore receive identical target scores. This is relevant both to regression performance and to the usefulness of the score for ranking. Table A.1 gives the category counts so that the realised distribution can be inspected independently of the generator's intended probabilities.

3.3 Experimental and Study Procedure

The training notebook reads the CSV and encodes service_category and highest_cert_level using separate LabelEncoder objects. The provider identifier and target-related columns are excluded from the prediction matrix. The target variable is credibility_score. The feature matrix is split with test_size set to 0.2 and random_state set to 42, giving 1,600 training records and 400 holdout records [12].

Random Forest and XGBoost are tuned on the training partition using GridSearchCV with cv set to three and scoring set to r2. Random Forest searches 324 parameter

combinations, corresponding to 972 fold fits. XGBoost searches 243 combinations, corresponding to 729 fold fits. Table A.2 preserves the searched parameter values. Both procedures refit the selected estimator on the training partition.

After tuning, the notebook predicts the holdout targets and prints RMSE and R squared. It creates actual-versus-predicted plots, residual plots and a learning-curve-style diagnostic. The latter repeatedly fits the selected estimators on fractions of the training partition and measures error on the same holdout set. At the final fraction of one, it refits on the entire training partition before subsequent feature-importance analysis and model saving.

The higher holdout R squared selects the model that is saved. This uses the test set for a model-selection decision, so the final holdout results should be treated as a comparative development result. A later study should reserve a further untouched set after selecting the algorithm and its parameters. The learning-curve diagnostic also reuses the holdout set and should not become an additional tuning mechanism.

3.3.1 Local Document Demonstration

The local inference procedure loads the saved XGBoost estimator and reconstructs the categorical encoders from the known category lists. It collects the provider's document paths, extracts available text, applies feature rules, constructs the ten-column prediction matrix and predicts a score. The result is clipped to the permitted range and mapped to a level.

The demonstration runs on four named example profiles in the notebook. Their results are reported under anonymous fixture labels in Chapter 6. A separate example processes the registration-document folder alone, showing how the model behaves with missing identity and other zero-valued inputs. That example is useful for identifying a training-to-inference distribution difference; it is not a validated assessment of a real provider.

3.3.2 Integration Demonstration

The integration procedure queries the providers node and selects records whose documents.status value is true. For each selected record, it downloads the document

groups into a temporary directory, creates features and a score payload, then updates result and status fields. A cleanup block removes the temporary directory after processing.

The notebook records one processed provider with five downloaded documents and a predicted score of 84.60. It records successful execution of the database update call. This supports a demonstration of the integration path. It does not establish sustained uptime, concurrent correctness or agreement between the raw document contents and an expert verification decision.

3.4 Measures and Evaluation Metrics

RMSE measures the square root of the mean squared difference between target and prediction. Its unit is credibility-score points. Lower RMSE indicates a closer approximation of the numerical labels on the evaluated records. R squared compares the residual sum of squares with the variation of the true targets. A value near one indicates that the predictions explain most of that target variation on the evaluated set.

$$RMSE = \sqrt{\frac{\Sigma\,(y_i - p_i)^2}{n}}$$

$$R^2 = 1 - \frac{\Sigma\,(y_i - p_i)^2}{\Sigma\,(y_i - \bar{y})^2}$$

Equations 3.2 and 3.3 Regression evaluation metrics

Here n is the number of evaluated records, y_i is the true target, p_i is its prediction and $\bar{y}$ is the mean true target.

These are regression measures. R squared of 0.994 must not be restated as 99.4% classification accuracy or a 99.4% probability that a provider is trustworthy. The project maps predictions to categories, but the training notebook does not report a level confusion matrix, precision, recall or macro F1. Those would be supplementary measures for the thresholded output.

For document extraction, a suitable future metric is exact agreement between an extracted field and a manually labelled reference. Experience durations could also be evaluated with absolute error, while document-presence and document-authenticity

decisions should have separate reference labels. Runtime should be measured independently for downloading, PDF extraction, OCR, feature interpretation, inference and database writing. These measurements were not reported in the existing notebook outputs.

3.5 Data Analysis

The analysis first compares the two recorded regression outcomes on the same holdout data. It then examines the target distribution, score saturation and constant features to interpret those outcomes. Model plots are used to understand fit behaviour rather than to substitute for numerical metrics. Explainability is considered as a diagnostic for the scoring model, not as proof of the validity of the scoring policy.

A supplementary offline dataset audit recomputes the generator's formula for all records. The calculated targets exactly match the stored credibility_score values. A deterministic implementation of the known rule therefore has RMSE zero and R squared one on these labels. A second supplementary baseline predicts the training-target mean for the 400 holdout records and obtains RMSE 14.698 and R squared approximately -0.0011. These baselines contextualise the learning task.

The exact formula baseline is particularly informative. It establishes that the experiment tests approximation of an available rule. A machine-learning model can provide a useful development exercise and a future architecture for independently labelled data, but the present results do not show that a learned model is necessary to calculate this policy.

3.6 Ethics Privacy Security and Responsible AI

The synthetic modelling dataset reduces the need to use personal provider records for initial algorithm development. Document materials and the live integration record nevertheless require careful handling because identity files, names and storage references can expose personal information. The thesis includes feature schemas and anonymised outputs without copying identity images, private keys, live storage links or full provider identifiers.

Responsible interpretation requires describing the score as a prototype decision-support indicator. Providers should be able to understand what evidence contributed to the result and correct extraction mistakes. A provider without formal certificates may still have substantial skill, so a documentation-based policy can disadvantage people whose professional experience is poorly documented. This risk should be examined with representative data and a review process.

The code's verification flags need additional controls before operational use. An uploaded file can be unreadable or unrelated to its declared type. The integration fallback can still set identity and registration indicators when a file exists. A successful scoring call also leads to verified flags being written. Authenticity should instead require evidence-specific validation and, where needed, human or authority review.

Credential management is another operational requirement. Administrative credentials should be held outside distributed source archives and accessed through an appropriate secret-management process. The uploaded component contains a service-account credential file; its contents are excluded from the thesis. Database permissions, retention arrangements, consent and access logging require review within the university and deployment context rather than an unsupported claim of legal compliance.

3.7 Reproducibility and Research Artefacts

The reproducibility artefacts are the dataset generator, CSV, training notebook, saved estimator, local inference notebook and integration notebook. Table 3.3 identifies the role of each non-sensitive artefact. The notebook outputs are useful historical records, but the repository does not include a pinned environment specification, separate saved encoders or a complete automated test report.

Table 3.3 Research artefacts and their roles

Artefact	Role	Evidence preserved
dataset/dataset-create.ipynb	Generate synthetic profiles	Seed settings and target formula
dataset/dataset.csv	Provide structured experimental data	2000 records and thirteen columns

Artefact	Role	Evidence preserved
model.ipynb	Tune compare and explain models	Recorded parameters metrics and plots
model/xgb_provider_credibility_model.pkl	Store selected estimator	XGBoost model file
inferance.ipynb	Process local document groups	Extraction rules and example outputs
integration.ipynb	Process platform documents	Download scoring and update logic

An independent rerun should record exact package versions, preserve the literal None qualification value and verify categorical encodings. CSV readers can interpret strings as missing values depending on configuration. A saved preprocessing pipeline would avoid reconstructing encoders independently in deployment. The final model, preprocessing definitions and schema should be versioned together.

For an offline rerun, generate or load the CSV, execute the modelling workflow and evaluate local fixture documents. Integration should first run against a test database or emulator with non-sensitive examples. A live administrative credential is not required to reproduce the synthetic regression comparison. Appendices C and D describe the current interfaces and a more complete verification procedure.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The component must accept a group of documents associated with one provider, extract available text and represent the relevant evidence in a fixed feature schema. The model must receive categorical values from its known vocabulary and numerical features in the order used during training. The score must be bounded between zero and one hundred and mapped to a documented level. These functional requirements connect the document and modelling parts of the system.

The integration must identify pending records, retrieve their documents and store the result under the correct provider. A result should retain both structured features and the prediction so that administrators can inspect the inputs behind the score. The evaluation timestamp provides a record of when the assessment was produced. It does not identify the model version in the current implementation.

Non-functional requirements include reproducibility, understandable feature semantics, controlled handling of personal files and safe behaviour when documents cannot be read. The existing implementation satisfies some requirements at demonstration level while others remain design requirements. Table 4.1 distinguishes them so that proposed safeguards are not presented as implemented capabilities.

Table 4.1 Requirements and implementation status

ID	Requirement	Current status
FR1	Read PDF and image evidence	Implemented with dependency and first-page limitations
FR2	Produce ten model inputs	Implemented using project extraction rules
FR3	Return bounded score and level	Implemented in prediction functions
FR4	Process local example providers	Four recorded fixture demonstrations
FR5	Store result in provider database	One recorded integration demonstration

ID	Requirement	Current status
NFR1	Preserve feature and model consistency	Feature order explicit encoders reconstructed
NFR2	Separate unreadable from unauthenticated	Not fully implemented
NFR3	Measure extraction accuracy	No labelled benchmark reported
NFR4	Audit retries and concurrent processing	Not demonstrated

4.1.1 Acceptance Criteria

For the numerical workflow, acceptance requires an input frame with the expected ten columns, a numerical output in the permitted range and consistent threshold mapping. At exactly sixty the category is Intermediate; at exactly eighty-five it is Professional. These criteria are deterministic and can be checked without claiming anything about the provider's real-world reliability.

For document processing, a stronger acceptance criterion would require every derived field to identify its evidence source and extraction outcome. A readable certificate should produce the expected supported category. An unreadable certificate should produce an explicit unresolved state rather than a confident qualification value. This stronger criterion is proposed for further development because the current fallback logic can assign a qualification based on uploaded document presence.

For database integration, acceptance requires that the update affects only the selected provider, preserves unrelated profile data and records any processing failure. A successful database write should follow adequate evidence checks rather than merely follow model prediction. Failure and retry behaviours should be tested before multiple workers or a production user base depend on the component.

4.2 Component Architecture Within the Complete Solution

The architecture has an offline model-development route and an online document-processing route. The development route creates the dataset, encodes predictors, tunes

models and stores the selected estimator. The processing route begins with provider documents, extracts features and calls that estimator. These routes meet at the feature schema and categorical mapping.

Figure 4.1 shows the component architecture. The provider platform supplies provider records and document references. The Python worker performs extraction and prediction, then returns a payload to Firebase. The platform can consume that result, but the supplied component does not demonstrate how other group modules use the score in booking, ranking or customer presentation.

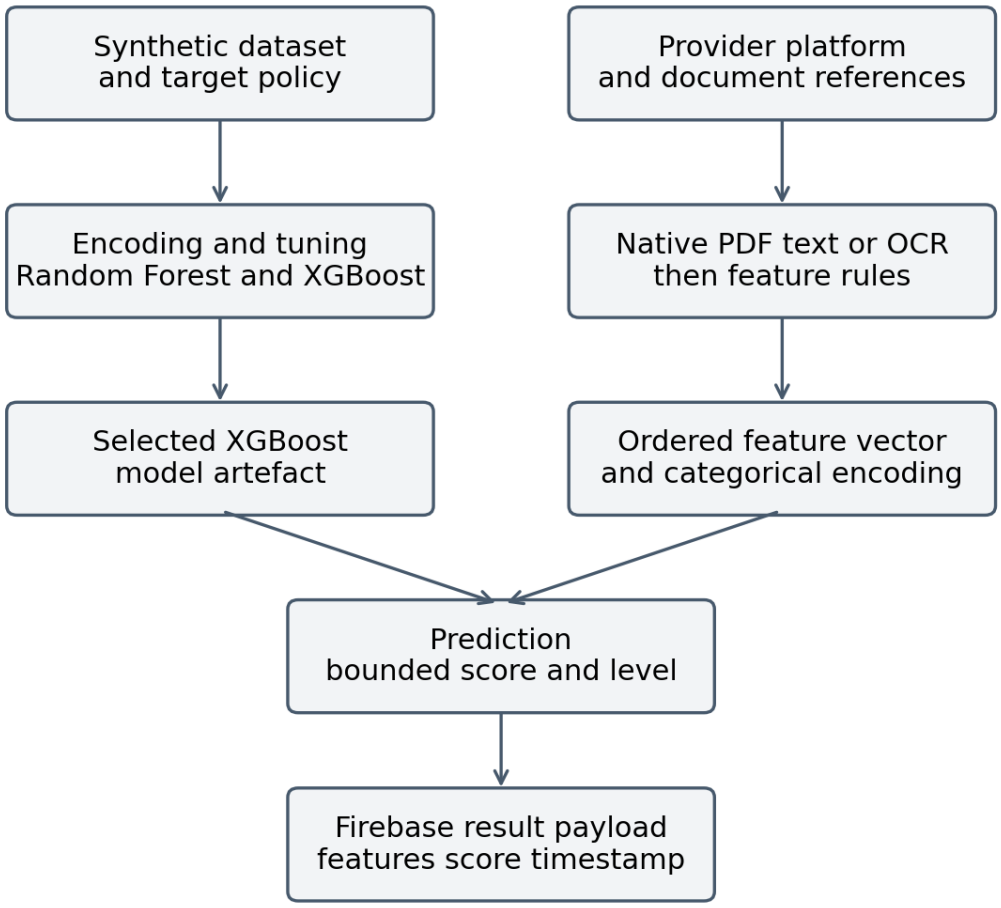


Figure 4.1 Architecture of the individual component

The local notebook is an important boundary in this design. It allows document processing without database downloads and updates. The integration notebook reuses the conceptual sequence but implements different rules for certain fields. This design

enables platform adaptation, while creating the possibility of disagreement between local and integrated results for equivalent evidence.

The database acts as the shared interface between the component and the rest of the platform. The update payload contains extractedFeatures, credibility data and processing flags. The extractedFeatures name expresses a useful intention, but some values are heuristics or fallbacks rather than directly extracted facts. Future schema design should preserve that distinction at field level.

4.3 Design Alternatives and Decisions

The simplest alternative is to compute the weighted target directly. Because the training labels are generated by a known formula, direct computation reproduces them exactly. It also makes each policy weight visible. The choice to compare machine-learning regressors is therefore justified as a prototype investigation and preparation for a future target learned from independent labels, rather than as a proven necessity for calculating the current score.

Random Forest and XGBoost provide two tree-ensemble approaches suitable for comparing structured predictions. A linear model is another possible benchmark, but the upper cap creates a change in the target response that an uncapped linear prediction would need to address. A capped weighted rule remains the most direct benchmark for this specific synthetic task.

For document extraction, native PDF text is attempted before OCR. This avoids rasterising a document when readable text is already present. Images require OCR if their text is to be used. A learned document model would offer a different extraction architecture, but the component lacks the labelled document dataset needed to evaluate that alternative in the same context.

The integration uses periodic polling. It is straightforward to inspect and can operate from a notebook, but it retrieves the providers collection before filtering pending records. An event-triggered or queue-based worker would offer a more scalable route. A protected claim or transaction would also reduce the chance that two workers process the same provider simultaneously.

4.3.1 Representation of Categories

The implementation uses LabelEncoder for service and certificate categories. This gives repeatable integer values when the category vocabulary and ordering are reproduced. It also gives an apparent numerical ordering to nominal service categories. Trees can split those codes, but the order itself has no professional meaning. Alternative representations include one-hot encoding or a model's supported categorical feature mechanism.

The certification policy ordering and the LabelEncoder ordering serve different purposes. The generator uses an explicit qualification list to assign points. LabelEncoder ordinarily assigns codes based on its fitted class ordering. A tree model can still approximate the mapping, but the two integers should not be confused. Appendix C documents this distinction for reproducibility.

Unknown categories are handled differently in the inference routes. The local encoder may raise an error when an unsupported value is supplied. The integrated normalisation function instead returns Electrician when it cannot recognise the input. A future design should return an unsupported or review-required state, preserving the original category for diagnosis.

4.4 Detailed Design

4.4.1 Document Acquisition and Text Extraction

Local inference receives a folder path, a single file path or a list of paths. A folder is recursively searched for selected file extensions. The integration receives document metadata organised into businessRegistration, certification, experienceProof, identityDocument and portfolioWork. It downloads recognised entries to a temporary directory and groups the resulting local paths by declared document type.

The text function first checks whether the path exists. For PDFs, it invokes pdftotext and uses the returned text when available. If that text is empty and both rasterisation and OCR dependencies are available, it renders the PDF at two hundred dots per inch and applies OCR to the first page. For supported images, it attempts image-to-text recognition. Errors are generally caught and converted into empty text.

Returning empty text simplifies downstream processing, but it removes information about the cause of failure. The caller cannot distinguish a missing executable, an unreadable scan, an unsupported format or a blank document. A structured extraction result should instead include status, page coverage, error information and recognised text. That would allow uncertainty to be carried into feature derivation and the platform display.

4.4.2 Feature Interpretation

The local identity rule looks for identity-related phrases or a number pattern compatible with the formats coded in the notebook. This detects a textual pattern; it does not confirm the document holder against an issuing authority. Business-registration detection similarly uses phrases such as business registration and registration number. The integrated route adds document-presence conditions, making its acceptance more permissive.

Certificate extraction counts documents with recognised qualification terms, assigns a category and gives an issuer-reputation heuristic. The local and integrated values differ. For example, the local NVQ Level 4 rule assigns 0.95 while the integrated rule assigns 0.92. Integration maps NVQ Level 5 evidence into NVQ Level 4 because the model does not support Level 5. Such transformations should be recorded as mappings rather than presented as exact recognition of the original qualification.

Experience extraction searches for integer year durations, including forms such as six (6) years, 5 years and recognised English number words. Within a document it takes the maximum detected year value. Across documents the extraction functions sum the detected years. This can overcount overlapping employment periods or duplicate letters. Date-range parsing and evidence deduplication would be necessary for a reliable total.

Portfolio count is based on recognised local portfolio indicators or files in the declared integrated portfolio group. Portfolio quality is assigned from text cues such as project descriptions and completed-work wording. There is no implemented computer-vision assessment of workmanship or image clarity. The feature should therefore be understood as a heuristic description of available portfolio documentation.

4.4.3 Prediction and Score Presentation

The feature dictionary is converted to a one-row DataFrame. The two categorical values are encoded, then the ten feature columns are selected in the training order. The saved estimator predicts a numerical value. The local and integrated functions constrain it to the interval from zero to one hundred and round it to two decimal places before assigning the level.

The level rule is score less than sixty for Beginner, score from sixty to less than eighty-five for Intermediate and score at least eighty-five for Professional. Because the inference functions apply thresholds after rounding, values very close to a boundary can change category due to rounding. A revised design should define whether levels use the raw score or displayed score and apply the same policy everywhere.

The result should be displayed with evidence coverage and a definition of the score. A number alone gives users no way to distinguish a model approximation from an authenticated professional credential. Provider correction and administrative review functions would make the system more usable when extraction is uncertain or the scoring policy does not reflect a provider's circumstances.

4.4.4 Database State and Result Payload

The integrated payload nests credibilityLevel, credibilityScore and lastEvaluatedAt under credibility within extractedFeatures. Other fields preserve counts and indicators. The stored service_category preserves the original platform category string, while the prediction input uses the normalised category. This supports display continuity but can make a categorisation error less visible unless the normalised value is also recorded.

After prediction, the update marks documents.status false, documents.verified true and the provider-level verified value true. The current worker therefore conflates completed processing with successful verification. A more precise state model would include pending, processing, extractionFailed, reviewRequired and completed, with authenticity decisions stored independently. Figure 4.2 presents that proposed state model and is explicitly a future design.

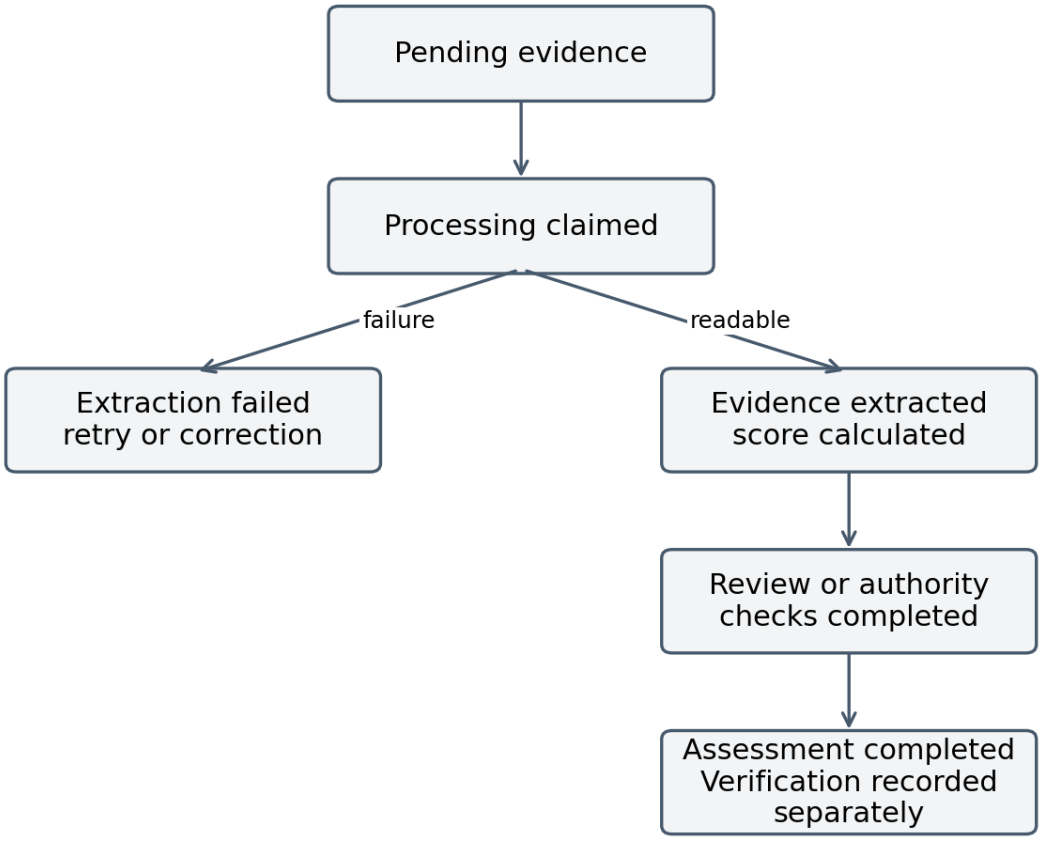


Figure 4.2 Proposed separation of processing and verification states

4.5 Standards and Professional Practice

Professional practice requires that the implementation, reported results and user-facing meaning agree. The thesis uses the recorded notebook metrics rather than estimated examples in descriptive documentation. It identifies heuristic fields by their implementation and avoids claiming that document presence proves authenticity. These choices preserve a defensible relationship between evidence and conclusions.

The component would benefit from a versioned input schema, pinned dependencies, saved preprocessing artefacts and clear error contracts. Model and encoder versions should be stored with each result. File paths should be constructed from controlled identifiers, and external downloads should be limited to permitted sources, formats and sizes. Administrative credentials should be separated from experimental distributions.

IEEE-style numerical citations are used for referenced publications and project artefacts. Equations identify the operational scoring and evaluation definitions. Tables and figures distinguish implemented architecture, observed outputs and proposed improvements. The prototype's limitations are part of the research result rather than hidden exceptions to its advertised behaviour.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The supplied component is organised as Jupyter notebooks and supporting files. Python provides the data-processing and model interfaces. pandas and NumPy support the tabular experiment. scikit-learn provides the Random Forest regressor, categorical encoding, splitting, parameter search and metrics. XGBoost provides the selected boosting estimator. Matplotlib and seaborn create model diagnostics, while SHAP is used for prediction explanations [12].

The document workflow uses Pillow, subprocess-based PDF extraction, optional pytesseract and optional pdf2image. Tesseract and Poppler are external dependencies. The recorded execution paths show a Windows-oriented setup. The code searches several Windows installation paths and otherwise relies on available executable names. Portable deployment would require configurable paths or an environment with the binaries explicitly installed.

The platform integration uses requests for document retrieval, firebase-admin for database operations, temporary directories for downloaded files and joblib for loading the estimator. The installation document lists package requirements and setup guidance, but its minimum versions are not an exact lock of the environment that produced the recorded metrics. Table 5.1 summarises the stack by function.

Table 5.1 Technology stack by function

Function	Technology	Role
Experimental data	Python NumPy pandas	Generate and prepare provider records
Model comparison	scikit-learn XGBoost	Fit tune and evaluate regressors
Diagnostics	Matplotlib seaborn SHAP	Plot fit behaviour and feature contributions
Model persistence	joblib	Store and load selected estimator
PDF extraction	Poppler pdftotext	Read embedded PDF text

Function	Technology	Role
Image recognition	Tesseract pytesseract Pillow	Extract text from images
Scanned PDF support	pdf2image	Rasterise pages for OCR
Platform integration	Firebase Admin requests	Read provider data download and update

5.2 Component Implementation

5.2.1 Dataset Generation

dataset-create.ipynb implements simulate_provider and generates two thousand records. Random service categories, qualification combinations and numerical values create a repeatable experimental table. The target formula is calculated within the same function that creates the input fields. The level is then derived from the score, and the resulting table is saved as CSV [11].

This construction provides complete numerical labels at low development cost. It also embeds the developer's policy assumptions directly into the training target. There is no independent annotation step, observed reliability outcome or unknown relationship to discover. The dataset is useful for checking model and integration plumbing while a representative empirical dataset remains a future requirement.

The CSV contains thirteen columns rather than the larger feature count suggested in some descriptive documentation. The predictor count is ten because the identifier, numerical target and level target are excluded. Preserving these counts in the thesis makes the experimental design easier to reproduce and avoids treating the level as an additional input that would leak target information.

5.2.2 Model Training and Selection

model.ipynb constructs the feature matrix, fits encoders and performs the split. Random Forest is searched across estimator count, depth, split size, leaf size and feature selection. XGBoost is searched across estimator count, depth, learning rate,

row subsampling and column subsampling. The selected settings are preserved in Table 5.2 [12].

Table 5.2 Recorded selected model parameters

Model	Parameter	Selected value
Random Forest	n_estimators	100
Random Forest	max_depth	12
Random Forest	min_samples_split	2
Random Forest	min_samples_leaf	1
Random Forest	max_features	None
XGBoost	n_estimators	300
XGBoost	max_depth	5
XGBoost	learning_rate	0.05
XGBoost	subsample	0.7
XGBoost	colsample_bytree	1.0

The recorded Random Forest best cross-validation score is 0.9828161484. The notebook prints XGBoost's selected parameters but does not print its best cross-validation score in the provided output. It would be incorrect to invent that value from the final holdout result. Chapter 6 therefore reports the values that are actually available.

The notebook selects the saved estimator using the model-comparison table's holdout R squared. XGBoost is saved as xgb_provider_credibility_model.pkl. The encoders are not included as separate saved files in the component. The inference notebooks rebuild their class vocabularies, which requires careful agreement with the training representation.

5.2.3 Diagnostics and Explainability

The training notebook creates predicted-versus-actual plots and residual plots for both models. These diagnostics help identify whether errors vary across score regions. It also constructs a training-fraction diagnostic and plots model feature importances. The plots are recorded outputs rather than a quantified statistical analysis of residual normality or uncertainty.

SHAP code constructs an explainer and creates global summary, dependence and local explanation views. No numerical ranking of feature contributions is printed in the supplied textual output. The thesis therefore does not assert that identity is the most influential feature. Because identity is constant in training, a learned response to changes in identity cannot be evaluated from these records.

An explanation should be read as a description of the model relative to its data and explanation background. In this case the model is learning a synthetic scoring rule. A strong contribution from experience would reflect that rule and its generated distribution, not an independent finding that experience causes a particular level of trustworthy service.

5.2.4 Local Inference

inferance.ipynb loads the saved model and defines extract_text_from_file, parse_service_category, parse_experience_years, extract_features_from_provider, prepare_ml_input, predict_credibility and evaluate_provider. The functions form an understandable sequence from files to result. evaluate_provider returns the extracted fields together with the numerical score, level and demonstration name [13].

Service category can be inferred from text or filenames. If several documents suggest categories, the first detected category is used. If none is found, the default is Electrician. File order can therefore influence the selected category in ambiguous cases. Explicit provider-declared categories and conflict reporting would be preferable to silent defaulting.

The local batch example lists the documents for four profiles and prints a structured summary. Another example passes a registration folder to the same extraction

function. This demonstrates interface flexibility, but the resulting feature vector includes values outside the generated training domain, such as identity zero and experience zero. Prediction remains possible, while its validity is not established by the training experiment.

5.3 Integration With the Other Components

The demonstrated integration boundary is the Firebase provider record. integration.ipynb selects pending records, processes the five document groups and writes extractedFeatures under the same provider. It also changes document and provider verification flags. The implementation shows how this individual component can make its results available to the wider platform without embedding the model into every client [14].

The integration normalises platform categories using a dictionary. Plural values such as Electricians map to Electrician. The category a/c services maps to AC Repair. The recorded example uses A/C, which is not a key in the normalisation dictionary. The actual function therefore defaults to Electrician for that input, while the stored result preserves A/C. This discrepancy is an integration limitation, not evidence that AC Repair was correctly supplied to the model.

The code accepts document lists represented as lists or dictionaries, which accommodates possible database structures. Each entry can contain fileUrl, fileName, format and fileId. Downloaded files are grouped by document type. The response timeout is thirty seconds. Failed downloads are skipped, but the processing routine can continue with an incomplete document set and still update verified flags.

No other group component source code is supplied within this individual ZIP. The thesis consequently describes the interface rather than assuming a completed customer-facing ranking, recommendation or booking integration. Platform acceptance testing should verify that the result is used with suitable labels, provenance and review controls.

5.4 Verification and Software Testing

The provided notebooks contain demonstrations and diagnostic outputs, but no complete automated test suite or formal acceptance-test report. Their evidence should be described as functional examples. A successful example shows that a path executed for its inputs; it does not establish comprehensive behaviour across unsupported categories, failed downloads or unreadable files.

Supplementary offline checks examine isolated helper functions without loading the model or connecting to Firebase. The experience parser returns six for six (6) years, five for 5 years and twelve for twelve years. It returns zero for 18 months. The category normaliser maps Electricians correctly and maps a/c services to AC Repair, but maps A/C and an unknown category to Electrician. Table 5.3 records these checks.

Table 5.3 Supplementary offline helper checks

Helper	Input	Returned value	Interpretation
Experience parser	six (6) years	6	Supported duration form
Experience parser	5 years	5	Supported numeric form
Experience parser	twelve years	12	Supported word form
Experience parser	18 months	0	Month duration unsupported
Category normaliser	Electricians	Electrician	Expected mapping
Category normaliser	a/c services	AC Repair	Expected mapping
Category normaliser	A/C	Electrician	Default hides an intended AC category
Category normaliser	unknown	Electrician	Unknown input silently defaulted

These checks confirm code behaviour rather than OCR quality. The input is already text, so the tests do not evaluate recognition from images. They also do not change the user's project files. A further suite should distinguish parser tests, document extraction

tests, model tests and database state tests, with expected results defined before execution.

5.4.1 Model and Feature Verification

The ten-column order should be checked against the saved estimator before inference. Numeric ranges should be validated, and unsupported categories should produce an explicit error or review state. Empty document groups should not automatically imply that a person is unskilled; they should indicate the absence of evidence needed for this assessment.

Boundary tests should check scores below, at and above sixty and eighty-five, as well as clipping at zero and one hundred. Threshold policy should be shared between dataset generation and inference. A score computed from continuous inputs can be rounded across a boundary, so tests should explicitly define the intended relation between displayed values and category assignment.

Duplicate evidence is another important test case. Repeating an experience letter or certificate can increase counts or summed years. A robust pipeline should recognise duplicate files or overlapping employment periods. The current code does not demonstrate this behaviour, so it is included in the proposed verification matrix in Appendix D.

5.4.2 Integration and Failure Verification

The integration worker should be tested when the download returns a non-success response, when no readable text is produced, when the model is unavailable and when the database update fails. Processing should preserve the pending or review state if evidence checks cannot be completed. A transaction or worker claim should be tested for simultaneous processing attempts.

The existing finally block provides useful temporary-file cleanup. However, cleanup is a resource-handling guarantee rather than an evidence-validation guarantee. A worker can clean its temporary directory and still write an unjustified verification status. Tests must therefore inspect the state transition and payload as well as successful execution and file deletion.

5.5 Deployment and Operation

The recorded workflow runs through notebook execution. The integration notebook offers a continuous polling function with a default interval of fifteen seconds. The function calls the scanning routine repeatedly and sleeps between calls. Its activation line is commented out, so the archive does not establish that a continuously running service was deployed.

For deployment, the worker should be converted into a controlled service with a versioned model, preprocessing artefacts and configuration. The application should separate model readiness from document-extraction readiness. A missing OCR dependency must be reported because it changes document coverage, even if native PDF extraction and prediction still function.

Monitoring should record processing duration, extraction failures, download failures, model version and review outcomes. Reprocessing should be triggered by new evidence or an explicitly approved policy/model change. A score should not be silently overwritten without retaining the reason for reevaluation. A future production design should also provide a retention policy for raw documents and extracted personal information.

These are deployment directions rather than completed operational claims. The current evidence supports a notebook prototype with one recorded integration update, not a measured production service-level objective or a commercial deployment.

6 RESULTS AND VALIDATION

6.1 Research Results

The recorded model comparison favours XGBoost over Random Forest on the selected synthetic holdout set. Random Forest obtains RMSE 1.569 and R squared 0.989. XGBoost obtains RMSE 1.141 and R squared 0.994. Table 6.1 reports these values as stored notebook outputs, rounded to three decimal places [12].

Table 6.1 Recorded holdout regression results

Model	RMSE in score points	R squared	Evidence status
Random Forest	1.569	0.989	Recorded notebook output
XGBoost	1.141	0.994	Recorded notebook output

Using the rounded values, XGBoost reduces RMSE by 0.428 score points relative to Random Forest, approximately 27.3%. The comparison is between two models on one generated dataset and one selected split. It does not provide an uncertainty interval or demonstrate that the difference persists across real provider populations. Figure 6.1 presents the comparison with labelled axes.

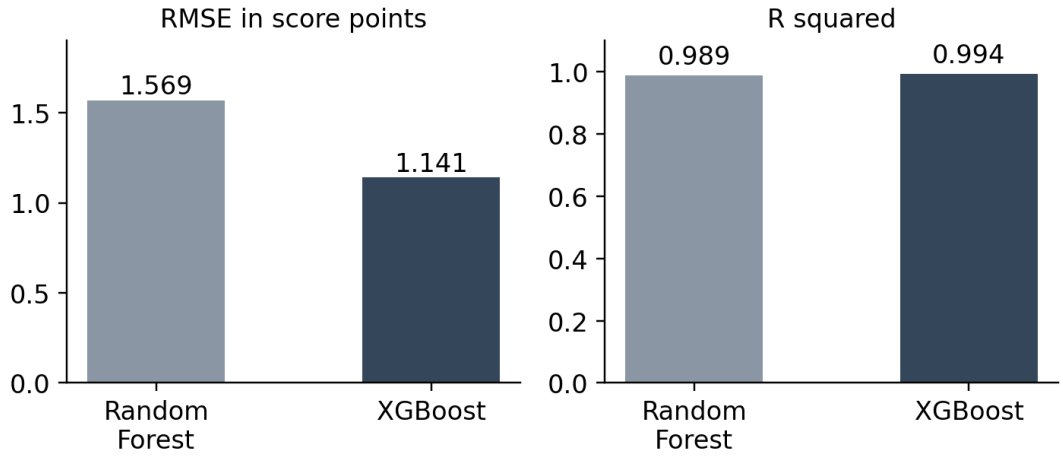


Figure 6.1 Recorded regression model comparison

The outcome answers RQ1 within the defined experiment: XGBoost is the better-performing recorded model for this synthetic policy-approximation task. Both models

closely reproduce the generated scores. This fit should be interpreted alongside the fact that the target is calculated from the inputs by a deterministic rule.

6.1.1 Supplementary Baseline Results

The offline audit reproduces every stored label using the formula in the generator. The maximum absolute difference is zero after applying the specified two-decimal rounding. This establishes a direct policy baseline with RMSE zero and R squared one. It is an exact rule calculation, not a learned regression model or evidence of perfect prediction of actual credibility.

The training-mean baseline uses the same seed-42 split and predicts a constant value from the training labels. Its holdout RMSE is 14.698 and its R squared is approximately -0.0011. Table 6.2 distinguishes these supplementary calculations from the recorded model outputs.

Table 6.2 Supplementary offline baselines

Baseline	Evaluation set	RMSE	R squared
Training target mean	400 holdout records	14.698	-0.0011
Exact generating formula	All 2000 records	0.000	1.0000

The ensemble models clearly improve on a constant predictor for the synthetic labels. The deterministic formula is nevertheless the appropriate reference for whether model learning improves calculation of the current rule. The available results do not establish such an improvement. A future independent target could change that comparison, but it would require a new empirical experiment.

6.1.2 Distribution and Feature Findings

All identity_verified values are one. Identity cannot explain variation between generated records because it does not vary. This contradicts an interpretation that training demonstrates the importance of identity verification. Its thirty-point

contribution exists in the generator's policy, but the model never observes a contrast between verified and unverified identities.

One-third of labels are capped at one hundred. The cap reduces distinctions between highly scored profiles and creates a large Professional group. A low overall regression error can therefore coexist with limited differentiation within that group. Tier-level and uncapped-region analyses would help determine whether useful differences are being obscured.

Four inputs are not direct terms in the label formula. Their model contributions may arise from generated correlations or approximation behaviour. These features should not be described as independently validated determinants of credibility. In particular, a portfolio count does not directly change the current label unless it is associated with another feature.

6.2 Product and Technical Results

The local document demonstration produces feature summaries and scores for four example provider profiles. Table 6.3 anonymises the fixture names while preserving their categories and recorded results. The profiles use the supplied document examples and are not a representative provider sample [13].

Table 6.3 Recorded local document inference examples

Case	Service	Years	Portfolios	Score	Level
A	Electrician	1	1	75.64	Intermediate
B	Carpenter	6	1	89.99	Professional
C	Carpenter	5	1	95.00	Professional
D	Electrician	9	2	100.00	Professional

All four summaries show identity_verified one, business_registered one and certification_count one. Fixtures A and B have NVQ Level 4 as the detected category. Fixtures C and D have Technical College. These values demonstrate the output structure and selected extraction behaviour. Without independent document annotations, they do not provide a field-accuracy percentage.

The training notebook includes two additional manually specified structured inference examples. The recorded Electrician example returns 94.51 and Professional; the Plumber example returns 74.85 and Intermediate [12]. These are different from the approximate illustrative values in the README. The thesis uses the notebook outputs as the experimental record rather than substituting estimates from descriptive documentation.

The registration-folder-only example produces identity zero, registration one and zero for the other evidence values, returning 50.94 and Beginner. This shows that inference can run with incomplete evidence. Since identity zero and experience zero were not represented in the generated training range, the prediction is not validated by the holdout experiment. It should be considered an input-domain warning rather than a reliable assessment of an undocumented provider.

6.3 User or Stakeholder Validation

No formal user study, supervisor rating dataset, customer survey or stakeholder interview result is recorded in the supplied component. Accordingly, this thesis does not report participant counts, satisfaction percentages or evidence that customers make better decisions using the score. Functional demonstration and user validation are different forms of evidence.

A suitable next validation would involve a defined review rubric and a representative set of anonymised documents. Independent reviewers could label readable fields and assess evidence adequacy before seeing the system's output. Agreement between reviewers should be measured because professional credibility may not have a single uncontested ground truth.

A separate usability study could evaluate whether customers and providers understand the score, its evidence coverage and its limitations. Useful tasks include identifying why a score was assigned, recognising a review-required state and correcting an extraction error. These proposed measures would test the interface meaning rather than merely whether participants like a numerical badge.

Provider validation should include applicants with limited formal documentation and applicants whose evidence is not in English. Otherwise, the evaluation may underrepresent people most affected by the scoring policy. The study should also examine whether a documented qualification is relevant to the declared service category, an association not currently enforced by the extraction rules.

6.4 Integration Results

The integration notebook records one pending provider processed using five documents. The resulting extractedFeatures includes identity one, business-registration one, certificate count one, Technical College, experience one year, reference count one, portfolio count one and portfolio heuristic 0.90. The credibility result is 84.60 and Intermediate, followed by a successful database-update message [14].

Table 6.4 summarises the recorded integration behaviour without exposing the provider's name, identifier or document URLs. The timestamp in the original result dates that recorded run to August 2026; it is not evidence that the database was rechecked during thesis preparation.

Table 6.4 Recorded integration demonstration

Item	Recorded outcome	Interpretation
Pending records selected	1	One demonstration record
Documents downloaded	5	Download stage executed
Score and level	84.60 and Intermediate	Model output stored in payload
Original category	A/C	Preserved in stored result
Normaliser behaviour for A/C	Electrician fallback	Offline helper check reveals mismatch
Database update call	Success message	Recorded execution of update path

The category mismatch constrains interpretation of this integration result. The stored category remains A/C, but the normaliser defaults that string to Electrician before

encoding. Consequently, the demonstrated score cannot be treated as an independently verified AC Repair result. This illustrates why a stored original value and a model input value both belong in an audit record.

The recorded environment reports pdf2image unavailable. Native text extraction and image OCR can still run, but the scanned-PDF fallback is unavailable in that execution. The demonstration therefore does not establish complete support for all scanned PDFs. The current worker also writes verification flags after prediction without an independent authenticity decision.

6.5 Summary of Findings

The evidence supports three concrete findings. First, the synthetic regression workflow executes and the recorded XGBoost model has lower holdout error than Random Forest. Second, local document examples produce structured features and score levels. Third, the integration notebook records a successful path from pending provider documents to a Firebase update.

The evidence also defines clear limits. The label policy is known and exactly reproducible without machine learning. Identity does not vary in training, score saturation is substantial and no extraction or stakeholder benchmark is reported. Category defaults and permissive verification logic can make a technically successful workflow produce a result with weaker evidential meaning than its field names suggest.

These findings support further development of the component while keeping the present claim at prototype level. They answer RQ2 and RQ3 as demonstrated functionality and RQ4 as identified constraints. They do not establish commercial readiness or validated prediction of actual provider reliability.

7 DISCUSSION

7.1 Interpretation of Findings

The regression results show that the selected features are sufficient for both tree ensembles to approximate the generated target. This is expected because the same features determine the target formula. XGBoost's lower error is useful when selecting between the two recorded models, but it should not be interpreted as discovering a new empirical definition of credibility.

The deterministic baseline clarifies the role of the model. When a platform wants to enforce a fixed weighted policy, it can calculate that policy directly. A learned estimator adds approximation error and model maintenance. It becomes more substantively justified if the target comes from independent assessments or service outcomes that cannot be described by a fixed known formula.

The document pipeline provides a more distinct engineering contribution than the synthetic prediction result alone. It connects files, feature interpretation and the provider database. Its quality depends on the fidelity of those transformations. Improving the model's error by a fraction of a point would have limited value if an unreadable identity image or a defaulted category is presented as confidently verified evidence.

The component should therefore be understood as a structured onboarding prototype. It can help organise evidence and supply a consistent policy indicator. The next stage should strengthen extraction and verification semantics before using score differences to make high-confidence decisions about real providers.

7.2 Comparison With Literature Baselines and Existing Solutions

The Random Forest and XGBoost choices follow established ensemble methods [2], [3]. The contribution is their application to the project's provider schema, not a modification of the algorithms. The experiment supplies a fairer comparison between these methods than unrelated benchmark values because both use the same dataset and split. However, model selection on the holdout set limits the strength of the final generalisation claim.

The reputation literature concerns histories of interactions and the information they convey [1]. This component uses onboarding documents and a generated policy. The two evidence sources could complement each other in a future platform, but they should not be treated as interchangeable. A certificate indicates a qualification claim; a transaction outcome indicates performance in a particular service interaction.

The current extraction is simpler than layout-aware or OCR-free document models [6], [7]. Its rules are inspectable, which is helpful for debugging a small supported vocabulary. Its weaknesses become visible when documents have unexpected wording, uncertain OCR, mixed categories or durations expressed as dates and months. The literature provides design alternatives, but no result in this thesis establishes which would perform best on a representative PAWAN document collection.

Manual review and the deterministic formula are essential future baselines. Manual review can assess evidence quality and relevance with a defined rubric. The formula provides an exact policy output. A learned model should be compared with both when its target becomes independently defined. That would reveal whether the model improves prediction and whether document automation preserves the evidence used by the reviewer.

7.3 Research Contribution

The individual research provides a reproducible synthetic scoring experiment, a recorded comparison of two regression approaches and an analysis of their validity boundary. The explicit schema, seed settings and target rule allow the modelling task to be examined. The supplementary audit strengthens the interpretation by demonstrating exact policy reproducibility and identifying distribution properties that affect performance.

The study also contributes specific questions for subsequent empirical evaluation. These include whether the policy matches expert judgement, whether category-specific qualification relevance matters, whether missing documents unfairly reduce opportunities and whether extraction errors are more influential than regression errors. These questions arise from observed implementation behaviour rather than an unsupported claim of universal novelty.

A broader research contribution would require independent labels and representative validation. The present thesis does not claim that its generated score is a scientifically established measure of trustworthiness. It instead documents what the prototype measures and what evidence would be needed to extend that claim.

7.4 Product Contribution

The product contribution is an end-to-end connection between provider evidence and a database result. The notebooks expose the processing steps in a form that can be reviewed, demonstrated and extended. Structured output preserves several intermediate fields rather than storing only a score, allowing administrators to inspect counts and categories.

The integration provides a practical interface for the wider platform. A provider record becomes a processing request, and extractedFeatures becomes the response. Temporary download directories and cleanup support local processing without permanently accumulating files in the worker's working directory. Score clipping and level mapping provide consistent output formatting.

The product's next improvements concern uncertainty and review. The schema should represent evidence source, extraction confidence, unsupported values and authenticity status. A review interface should expose those fields and allow corrections. This would make the numerical result more accountable and would reduce the chance that an optimistic fallback becomes a misleading platform badge.

7.5 Limitations and Threats to Validity

7.5.1 Construct Validity

Construct validity concerns whether the measured score represents the concept claimed. Here, the target is a weighted document and experience policy. The weights are not validated against independently observed reliability. Issuer reputation and portfolio quality are generated or assigned heuristically. The title credibility must therefore be accompanied by an operational definition.

Qualification categories also lack a validated ordinal scale. The explicit list gives Vocational more points than Technical College and NVQ Level 4. The resulting hierarchy should be reviewed with domain expertise rather than assumed from the code. The Professional category is a threshold label, not recognition by a professional body.

7.5.2 Internal and Statistical Validity

The experiment uses one random split and no reported repeated-run uncertainty. Hyperparameters are tuned on training folds, but the model family is selected using holdout R squared. The holdout is also used repeatedly for the training-fraction plot. These development decisions should be separated from an untouched final evaluation in a future study.

The label has no stochastic measurement noise because it is calculated from the inputs. This makes high fit performance easier than a real-world prediction problem with inconsistent reviews and imperfect evidence. The cap also produces many identical targets. Overall RMSE can hide different errors in the small Beginner group and near category boundaries.

7.5.3 External Validity

The simulated ranges do not cover all inference conditions. All training identities are one, experience is at least half a year and portfolio quality is at least 0.30. Local inference can create zeros for those fields. Real providers may also have more than three certificates, more than fifteen years of experience or categories outside the ten supported values.

The sample documents do not establish representative variation in scanning conditions, languages or document templates. Four local demonstrations and one database update are too limited to support a claim about operational accuracy. Real-world validation must include incomplete, contradictory and unreadable evidence as well as clear examples.

7.5.4 Software and Integration Validity

The local and integrated extraction rules differ. Document-presence fallbacks, qualification remapping and category defaults can change model inputs independently of the provider's evidence. The A/C normalisation mismatch is a concrete example. Stored original categories do not reveal the encoded input, so inspection of result payloads alone can miss the issue.

The worker can update verified flags after incomplete downloads or weak extraction. No independent authenticity check is demonstrated. Concurrent processing, retry semantics and failure states are not fully specified. These are material limitations because the platform may interpret a successful write as a successful verification decision.

7.5.5 Reproducibility

Exact dependency versions and separate encoders are not preserved in the archive. Notebook execution order can mutate fitted estimators, including the training-fraction diagnostic before saving. CSV interpretation of the literal None value also requires care. A consolidated pipeline, environment lock and explicit artefact versions would improve repeatability.

The recorded model metrics are not newly retrained results from thesis preparation. The supplementary checks are limited to CSV calculations and pure functions. The thesis preserves that distinction so that a reader can evaluate the provenance of each finding rather than assuming that all results came from one controlled run.

7.6 Implications

The immediate development implication is to fix evidence semantics before expanding numerical scoring. Identity and registration indicators should express what the system actually checked. Unreadable evidence should remain unresolved. A score should not trigger a verified badge unless the verification criteria have independently been satisfied.

The research implication is to choose an independent target before making stronger machine-learning claims. Expert assessments, authenticated evidence or carefully defined service outcomes could provide a target, but each introduces its own biases and measurement problems. The study design must explain why the target reflects the intended use and how it is separated from model inputs.

For the user experience, the score should accompany an explanation of evidence coverage, relevant qualifications and review options. Providers should be able to correct facts. Customers should understand that a score is a decision-support summary with defined limits. This would make the component more useful than presenting an unexplained number as a universal judgement of a person's credibility.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The proposed value is assistance with organising and reviewing provider onboarding evidence. Platform administrators are the primary operational users because they must decide whether an application contains sufficient evidence. Providers could benefit from seeing which submitted items were recognised and which require correction. Customers could receive a clearer summary after the platform establishes a defensible interpretation of the score.

The potential benefit is more consistent handling of documents and reduced repetitive reading. The current prototype does not quantify time savings, customer conversion or reductions in disputed service bookings. These remain hypotheses for a pilot. A commercial claim should be based on measured comparison with the existing review workflow rather than the regression benchmark alone.

The initial target deployment is a controlled service-marketplace environment with a limited set of document categories and service types. Starting with a narrow supported scope would allow the team to label documents, resolve category mappings and review extraction failures. Expansion should follow evidence that the process works for the added services and documents.

8.2 Market and Competitive Position

The component can be positioned as an onboarding evidence processor connected to a service platform. Its distinction is the combination of document grouping, structured professional features and an integrated result payload. Its main competitive alternatives are manual review, simple document checklists and direct policy scoring. No current pricing or market-share comparison is established in this thesis.

The deterministic rule is a serious alternative because it calculates the present policy exactly. A commercial product should justify any use of machine learning through an improved empirical target or a demonstrated operational benefit. More complex modelling should not be presented as inherently more valuable than a transparent rule.

Competitive usefulness would depend on accurate extraction, review workflows, reliable integration and clear explanations. These are the areas where the prototype needs stronger evidence. Claims about fraud prevention, customer safety or improved provider reliability should be withheld until tested against independent outcomes.

8.3 Business or Sustainability Model

A proposed model is to offer document processing as a service integrated into a marketplace's onboarding workflow. A platform could fund it through its general operating budget or a subscription arrangement. Charging for processing volume is another possibility, but pricing should avoid rewarding false verification or tying a provider's credibility score to payment.

The operating costs would include document storage, downloading, OCR processing, database operations, model maintenance and human review of uncertain cases. Human review is likely to remain necessary for authenticity and ambiguous evidence. The project does not provide a measured cost model, so no revenue, margin or return-on-investment figure is claimed.

Sustainability should be assessed by the cost per completed review, the rate of unresolved cases and the quality of correction handling. A score that is inexpensive to compute but frequently wrong could create larger downstream costs. A pilot should compare review duration and error rates against the baseline administrative process.

8.4 Scalability Intellectual Property and Compliance

Scalability requires changing the current full-collection polling design when provider volume increases. A queue or event-driven worker can process targeted requests. Workers should claim jobs, record attempts and distinguish retryable failures from review cases. OCR resources can then be allocated independently from the lightweight regression prediction stage.

The project's own code, configuration and documentation need a clear ownership and licensing record. Third-party libraries must be used according to their licences. The open-source availability of an OCR engine does not remove responsibilities concerning the personal documents processed through it. Model artefacts, generated

datasets and uploaded identity documents have different retention and distribution requirements.

A deployment should review consent, access controls, document retention and the university or organisation's applicable data-handling requirements. The prototype archive alone cannot establish compliance with all legal or institutional obligations. A recorded technical update is similarly insufficient to justify that documents have been authenticated. These matters should be addressed before offering a formal verification service.

The inclusion of an administrative credential in a development archive also shows why release packaging must be separated from deployment configuration. Credentials and personal document access must not be bundled with public code or demonstration reports. Operational configuration should remain under the control of the authorised platform administrator.

8.5 Contribution to the Sustainable Development Goals

The proposed contribution aligns most directly with Sustainable Development Goal 8, particularly its attention to entrepreneurship, productive activity and decent work [10]. A clearer onboarding process could help skilled providers present their qualifications and experience. This is a potential pathway to benefit, not a measured impact of the prototype.

That benefit depends on inclusive design. A scoring system that heavily rewards formal documents can create barriers for competent workers who have limited access to certification or registration. Alternative evidence, transparent review and correction opportunities would help address this issue. The platform should evaluate how results differ by documentation availability rather than assuming that a common formula guarantees fairness.

Environmental sustainability can be considered through restrained document retention, avoiding unnecessary repeated OCR and processing only records that need reevaluation. These measures would also reduce operating cost. No energy or carbon

measurement was performed, so the thesis does not claim a quantified environmental benefit.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This individual component develops a provider scoring prototype that connects document-derived features to regression prediction and Firebase integration. The supplied implementation contains a synthetic dataset generator, a model-comparison notebook, a saved XGBoost estimator, local inference and a platform processing worker. Together they demonstrate the technical sequence from evidence files to a structured provider result.

On the selected synthetic holdout set, XGBoost records RMSE 1.141 and R squared 0.994, compared with Random Forest's RMSE 1.569 and R squared 0.989. XGBoost is therefore the better of the two recorded regressors for approximating the project's policy labels. Local demonstrations show four document groups producing features and score categories, and the integration notebook records one successful provider update.

The meaning of these results is limited by label construction and extraction behaviour. The target is a deterministic weighted rule that can be computed exactly, identity is constant in training and many scores are capped at one hundred. The prototype does not validate actual reliability or authenticate documents against authoritative records. Its defensible outcome is an implemented and evaluated onboarding-scoring prototype, with clearly identified requirements for further validation.

9.2 Achievement of Objectives

The feature-schema and dataset objective is achieved for a synthetic experimental setting. The CSV contains two thousand records with ten predictors and two target-related fields. The model-comparison objective is achieved through the recorded Random Forest and XGBoost tuning and holdout results. The analysis identifies XGBoost as the selected model within that setting.

The document-extraction objective is achieved at demonstration level for the supplied materials. The functions process native PDF text and supported images, derive features and return scores. Comprehensive extraction accuracy is not established. Scanned

multi-page documents, month-based durations and evidence authenticity remain outside the demonstrated capability.

The integration objective is achieved through a recorded provider-processing and update path. Its correctness is constrained by category defaults and verification fallbacks. The broader validation objective is partially achieved: software behaviour and benchmark limits are examined, while independent stakeholder and provider-outcome validation remain future work. Table 9.1 presents the objective-level assessment.

Table 9.1 Achievement of individual objectives

Objective	Outcome	Limit
Define schema and dataset	Achieved for synthetic data	No representative empirical sample
Compare regression models	Achieved in recorded experiment	One split with synthetic labels
Extract document features	Demonstrated on fixtures	No labelled extraction benchmark
Map score to levels	Implemented	Policy categories need interpretation
Integrate with Firebase	Recorded successful update	Failure states and mappings need revision
Validate real credibility	Not established	Requires independent target and study

9.3 Individual Research and Product Contributions

The individual research contribution is the documented scoring experiment and its critical interpretation. It explains what the models learn, how they are compared and why synthetic fit cannot be generalised directly to provider trustworthiness. The supplementary baseline calculations and helper checks provide specific evidence for improving the experimental design.

The individual product contribution is the connected processing workflow: document text extraction, rule-based feature derivation, model input preparation, score

presentation and database output. The component creates a concrete interface for the platform while retaining intermediate feature values. These functions provide a base for subsequent extraction and operational improvements.

The thesis confines attribution to this individual component. The final university submission should use the group's agreed project title, student details and contribution allocation. The technical claims should remain tied to reproducible artefacts and measured results rather than broader platform capabilities that are not demonstrated here.

9.4 Future Directions

The first priority is to separate processing completion from verification. Document-presence fallbacks should be replaced by explicit extraction and review states. Identity and registration authenticity should be assessed independently of whether a file was uploaded. Incomplete downloads or unreadable documents should retain a failure or review status rather than lead automatically to verified flags.

The second priority is to unify local and integrated preprocessing. The service normalisation dictionary should handle A/C correctly and reject unsupported categories explicitly. Certificate mappings and heuristic values should follow a single documented policy. Encoders, schema and model should be saved together, with their versions recorded in every result.

The third priority is a representative document benchmark. It should contain manually labelled fields, multiple formats, different scan qualities, multi-page documents and relevant languages. Evaluation should measure extraction agreement and its effect on the final score. Experience parsing should handle dates and months, and duplicate or overlapping evidence should not inflate years or counts.

The fourth priority is an independently justified target. Domain experts could first review the scoring rubric, after which a controlled study could compare policy scores, expert judgements and defined service outcomes. The deterministic formula and manual review should remain baselines. Repeated or nested evaluation should reserve an untouched final dataset and report uncertainty and subgroup performance.

The fifth priority is operational testing and a controlled pilot. A queue-based worker, safe credential management, retries, audit trails and provider correction should be tested before broader rollout. Pilot measurements should include processing duration, unresolved cases, correction frequency and agreement with reviewers. Only then can the project assess whether the component improves onboarding quality and supports useful marketplace decisions.

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GLOSSARY

Credibility score. The numerical output defined by the project's evidence and experience policy. In the present prototype, its training target is calculated from a weighted formula rather than independent service outcomes.

Credibility level. A presentation category derived from the numerical score using the thresholds sixty and eighty-five. The categories are not official professional qualifications.

Evidence provenance. Information showing which document or processing rule produced a feature value and when it was obtained.

Holdout set. Records excluded from fitting and used to examine prediction behaviour. In this project the same holdout is also used to choose the saved model family.

Identity indicator. The feature called identity_verified in the code. Its actual value is based on recognition or presence rules and must be distinguished from authority-level authentication.

Policy baseline. A direct computation of a defined scoring rule, used to compare the learned approximation with the rule itself.

Score saturation. The condition in which multiple different profiles receive the maximum target score because the calculated value is capped.

Synthetic dataset. Generated experimental records used to develop and test a workflow. These records are not a representative sample of observed service-provider outcomes.

Verification. A decision that specified evidence checks have been satisfied. It should be distinguished from uploading a file, extracting text and completing a prediction.

APPENDIX A DATASET AND EXPERIMENT DETAILS

A.1 Service Category Distribution

Table A.1 gives the realised category counts in the CSV. Random selection produces approximate distribution across categories, not exact equality. All two thousand records are included in this summary.

Table A.1 Service category counts

Category	Records	Percentage
Electrician	179	8.95
Plumber	215	10.75
Carpenter	212	10.60
Mason	204	10.20
Painter	177	8.85
AC Repair	213	10.65
Refrigerator Repair	201	10.05
Washing Machine Repair	195	9.75
TV and Electronics Repair	216	10.80
Mobile Repair	188	9.40

A.2 Hyperparameter Search Space

Table A.2 preserves the implemented search values from model.ipynb. The Random Forest max_features list is sqrt, log2 and None, which differs from an older descriptive README that lists auto. The notebook is the authoritative implementation record for this comparison.

Table A.2 Implemented parameter grids

Model	Parameter	Search values
Random Forest	n_estimators	100 200 300
Random Forest	max_depth	5 8 12 None
Random Forest	min_samples_split	2 5 10
Random Forest	min_samples_leaf	1 2 4
Random Forest	max_features	sqrt log2 None
XGBoost	n_estimators	100 200 300
XGBoost	max_depth	3 5 7
XGBoost	learning_rate	0.01 0.05 0.1
XGBoost	subsample	0.7 0.8 1.0
XGBoost	colsample_bytree	0.7 0.8 1.0

A.3 Numerical Audit Procedure

The supplementary audit reads the CSV with the literal None string retained as a category. It maps qualification categories to the generator's list indices, calculates the weighted sum, caps it at one hundred and rounds to two decimals. The recalculated values match the stored labels exactly. This audit does not load the pickle model or run any database operation.

For the constant baseline, the record indices are split using test_size 0.2 and random_state 42. The mean of the training labels is repeated as the prediction for the four hundred holdout records. RMSE and R squared are then calculated against their stored labels. The result is a baseline of target variation, not an alternative credibility definition.

A.4 Evidence Classification

Recorded outputs are the original notebook results from model training, local inference and integration. Supplementary audit outputs are calculations and helper checks

performed during thesis preparation. Proposed evaluations are procedures described for future work. Keeping these evidence classes separate prevents future test plans from being mistaken for completed experiments.

The model-comparison results have not been reproduced by a fresh end-to-end retraining in this preparation workflow. The saved estimator exists in the archive, but loading and retraining it would require an appropriate XGBoost environment and consistent preprocessing. Its presence does not resolve uncertainties about exact environment versions or execution history.

APPENDIX B ACADEMIC INTEGRITY AND AI USE DISCLOSURE

B.1 Assistance Used in Preparing This Thesis

ChatGPT with Codex assistance was used to inspect the supplied thesis guide, formatting guidance and individual component archive; organise and draft English thesis text; locate and check technical references; prepare diagrams and tables; calculate supplementary dataset summaries and deterministic baselines; inspect selected pure helper functions; and format and render the Word document for layout review.

Representative instructions were to prepare an individual thesis from the student's component ZIP and apply the uploaded content guide and formatting standards. The assistance produced drafting and document-preparation material. The project's original notebooks and their recorded outputs remain the basis for the reported historical model and integration results.

No claim is made that AI assistance collected independent provider outcomes, recruited participants, authenticated identity documents, interviewed stakeholders or performed a new live database verification. The supplementary offline calculations are identified in Chapters 3, 5 and 6. They should be reviewed and, where required by the module, rerun and documented by the candidate.

B.2 Candidate Review and Responsibility

Before signing the declaration, the candidate must check the technical account against their work, confirm the individual contribution allocation, verify the cited sources and review the supplementary calculations. The candidate must add any other AI or external assistance used in the original development, including coding, dataset design or analysis assistance not documented here.

The final disclosure should follow the module's actual policy. This appendix records the assistance visible in preparation of this document and should not be interpreted as

a complete history of every tool used during the project. The candidate remains responsible for the final claims, originality, code, evidence and interpretation.

B.3 Details Requiring Candidate Completion

The title page requires the official group project title, candidate's full name and student ID, group ID, degree, specialisation if applicable, department and submission month and year. The declaration requires candidate and supervisor names, dates and signatures. Acknowledgements require confirmation of the actual supervision, technical assistance and any funding.

Any additional experimental results should be added only with their procedure, dataset, environment and provenance. A stakeholder-validation section should remain as a proposed study until genuine data and approval requirements have been satisfied. There is no reason to invent missing results to make the document appear complete.

B.4 Use of the Supplied Guides

The formatting follows the supplied institutional guidance for A4 pages, Times New Roman, chapter and section sizes, line spacing, margins and pagination [15]. The nine-chapter body follows the content guide's table of contents, including commercialisation and sustainability [16]. The body of that guide skips the commercialisation chapter in its later heading list; this document retains the complete nine-chapter structure given in its contents.

The institutional formatting document mentions a minimum fifty-page dissertation and general chapter allocations. It also discusses group and individual reports together. The candidate should confirm any individual-component length requirement with the module supervisor. The academic substance should follow the individual research evidence rather than be expanded with invented experiments.

APPENDIX C TECHNICAL INTERFACES AND WORKED EXAMPLES

C.1 Exact Model Feature Order

The model input order is service_category_enc, identity_verified, certification_count, highest_cert_level_enc, cert_issuer_reputation, business_registered, experience_years, experience_reference_count, portfolio_quality_score and portfolio_count. Provider identifiers, names, credibility_score and credibility_level are excluded from the prediction input.

Keeping the order fixed is necessary when preparing a new one-row DataFrame. The schema should also define data types and allowed ranges. A provider identifier is preserved for database association but is not a predictor. Including the level target as an input would leak information derived from the numerical target.

C.2 Categorical Encoding and Policy Ordering

The generator's qualification indices are None zero, NVQ Level 3 one, NVQ Level 4 two, Technical College three and Vocational four. These values determine the five-point qualification term. They express the developer's scoring policy.

With the literal category strings retained, LabelEncoder's sorted qualification vocabulary is NVQ Level 3, NVQ Level 4, None, Technical College and Vocational. The resulting model codes are a separate representation. These should be saved from training rather than assumed from the policy index. CSV missing-value handling must be consistent before fitting or reconstructing the encoder.

The service vocabulary is AC Repair, Carpenter, Electrician, Mason, Mobile Repair, Painter, Plumber, Refrigerator Repair, TV & Electronics Repair and Washing Machine Repair in the sorted encoding order. Values from the platform should map explicitly into these classes. A stored original category should be accompanied by the normalised category actually used by the model.

C.3 Worked Policy Calculation

Consider the structured Electrician example in model.ipynb: identity one, two certificates, NVQ Level 4, registered business, five years of experience and portfolio quality 0.85. Under the target formula, the terms are thirty, sixteen, ten, fifteen, fifteen and 8.5. Their sum is 94.50. The recorded learned-model prediction is 94.51, illustrating approximation of the policy for this input.

The Plumber example has identity one, one certificate, Vocational, no business registration, three years of experience and portfolio quality 0.60. Its formula terms are thirty, eight, twenty, zero, nine and six, summing to 73.00. The recorded learned prediction is 74.85. The policy and learned prediction therefore agree on Intermediate but differ in their numerical output.

These examples explain how the component works. They do not show that the resulting values are independently correct measures of the example providers' professional reliability. The distinction between policy score and model prediction should remain visible when discussing them in a viva or demonstration.

C.4 Integrated Result Structure

The result is written beneath providers/{provider_id}/extractedFeatures. It contains the numerical and categorical feature fields and a credibility object with credibilityLevel, credibilityScore and lastEvaluatedAt. The current payload preserves the original category string in service_category. The paths also updated are documents/status, documents/verified and provider-level verified.

A future result schema should add modelVersion, preprocessingVersion, normalisedServiceCategory, evidenceSources, extractionStatus and reviewStatus. These additions are proposed interface fields, not claims that they already exist in the supplied code. They would support debugging, audit and correction without exposing private document text to all platform users.

C.5 Data Retention and Redaction

The thesis deliberately omits identity images, full live-provider identifiers, access-bearing file URLs and service-account secret contents. The demonstration records use anonymous fixture labels. The archive should be handled as development material containing sensitive configuration rather than as a public distribution package.

When creating further demonstrations, use fabricated or consented anonymised documents and a test database. A report should preserve feature values needed to understand the experiment while minimising personal information. Raw OCR text should be shared only when it is necessary and appropriately redacted.

APPENDIX D PROPOSED VALIDATION INSTRUMENTS

D.1 Document Annotation Protocol

A document benchmark should assign each file a document identifier, declared group, actual document type, language, format, page count and legibility category. Reviewers should label the fields relevant to that type, such as certificate level or experience duration. They should separately record whether authenticity can be established from the available evidence.

At least two reviewers should label ambiguous cases using a shared rubric. Differences should be reconciled or preserved as uncertainty rather than silently converted into a single confident truth. Reviewers should not see the automated score before labelling fields because that could influence their decisions. The benchmark should include readable and unreadable evidence in realistic proportions.

The dataset split should keep documents from the same provider or template family together where appropriate. Otherwise, closely related templates could appear in both development and evaluation sets. A held-out test should contain sufficient examples of the service categories and evidence conditions for which deployment claims are intended.

D.2 Proposed Test Matrix

Table D.1 lists high-priority tests to run before a wider platform pilot. The tests are proposed and have not all been executed. Their expected behaviour describes a strengthened design, which can differ from the current prototype's behaviour.

Table D.1 Proposed validation and failure tests

Case	Input condition	Expected strengthened behaviour
Native PDF	Readable embedded certificate text	Extract and retain evidence source
Scanned PDF	Relevant field on page two	Process relevant pages or report coverage limit
Unreadable identity	Blurred image with no usable text	Mark extraction unresolved avoid verification

Case	Input condition	Expected strengthened behaviour
Unsupported service	Category outside fitted vocabulary	Reject or require review preserve original value
AC category alias	A/C supplied by platform	Normalise to AC Repair explicitly
Month duration	Experience expressed as 18 months	Parse supported duration or require review
Duplicate evidence	Same letter submitted twice	Avoid double-counting duration
Conflicting evidence	Documents name different people	Flag conflict for review
Download failure	Missing or non-success response	Preserve failure state avoid verified flags
Concurrent workers	Same pending provider selected twice	Claim job once or perform safe idempotent update
Threshold boundary	Values close to sixty or eighty-five	Apply documented shared rounding policy
Model change	New estimator or encoders deployed	Store version and reason for reevaluation

D.3 Stakeholder Review Questions

For an administrator, ask whether the extracted fields match the documents, whether the system identifies uncertain evidence and whether the result saves time relative to the current process. Ask the reviewer to identify any field that could cause an incorrect verification decision. Record corrections and the time required for each case.

For a provider, ask whether the explanation accurately describes their evidence, whether they understand the difference between score and verification and whether they can identify how to correct an extraction error. Include applicants with different levels of formal documentation. Participation, consent and data handling should be arranged through the applicable study process.

For a customer, ask what conclusions they draw from the score and level. Check whether they interpret Professional as an official credential or verified guarantee. A

useful interface should communicate the intended evidence summary without encouraging stronger assumptions than the system supports.

D.4 Pilot Reporting Template

A pilot report should state the number and composition of evaluated cases, the reviewer rubric, data split, software versions and extraction coverage. Report exact field agreement, score error against the chosen reference, category confusion and processing duration. State unresolved cases separately from successful extractions.

For platform operation, report download failures, retry counts, review-required decisions and successful writes. For usability, report task success and correction behaviour rather than only general satisfaction. Every result should identify whether it is measured, derived or qualitative. This would produce evidence that can support a later revision of Chapters 6 and 7 without overstating what the prototype already demonstrates.